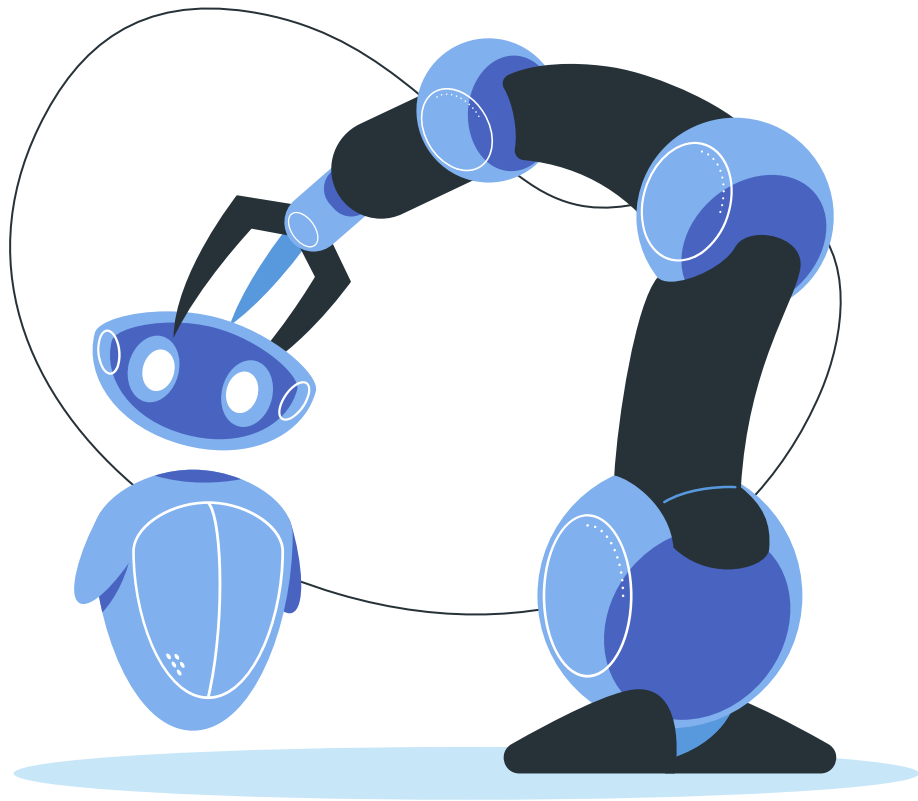


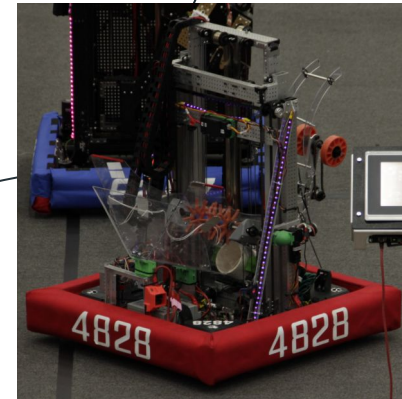
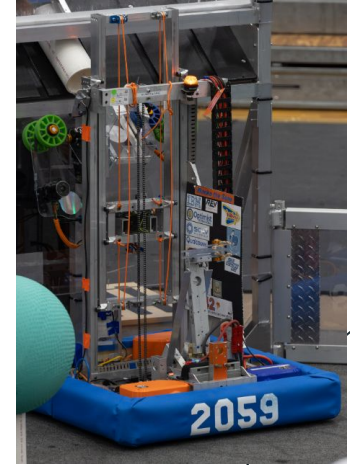
FRC Team Success Predictions

Ryan Fortner
Evan Lee
Dhruv Potnis



What is FRC?

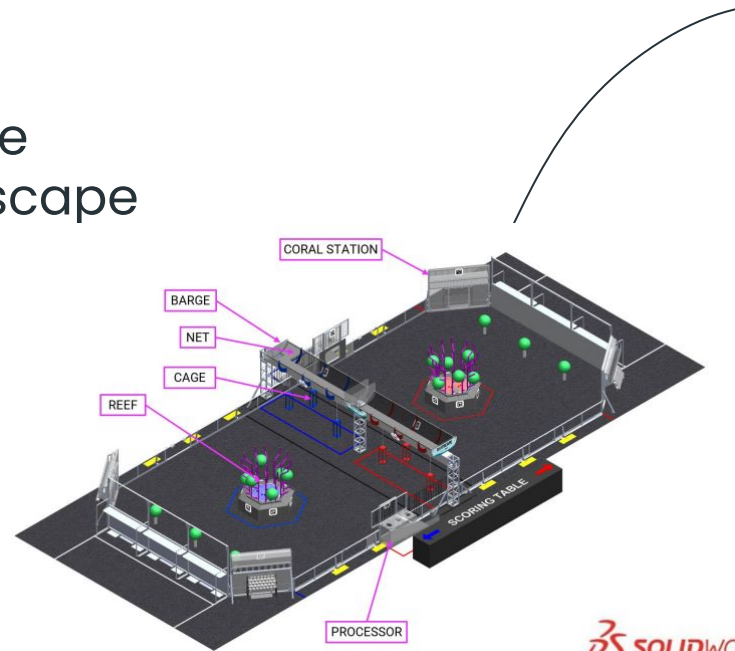
- FRC is a high school robotics challenge where teams build robots to interact and score game pieces
- Each match has two sides (alliances)
 - 3 teams on each alliance
- Around 60 qualification matches where alliances are randomly determined
 - Then top 8 ranked teams pick their own alliances for playoffs (double elimination bracket)
- Three core stages in a match:
 - Autonomous
 - Teleoperated
 - Endgame
- Accurate success prediction = strategic advantage
 - Most teams have “scouting” subteams
 - Using statistical analysis = Less human error, less people needed



Data Source – Overview

Statbotics.io, a commonly used FRC data analytics website. Aggregated from FIRST's official API, so data is highly trusted.

- Teams denoted by number and nickname
- Performance from the 2025 season, Reefscape
 - Pick-and-place game
 - Coral (PVC pipe)
 - Placed on a “reef” to score
 - Algae (kickball)
 - Numerous scoring options



Data Source – Headers

Header	Type	Description
team	String	Name of the team
total_epa, auto_epa, teleop_epa, endgame_epa	double	Expected points added in a different points of the match
rp_1_epa, rp_2_epa, rp_3_epa	double	Likelihood of getting Auto, Coral, or Barge ranking points, respectively
record	String	Formatted string, [Wins]-[Losses]-[Ties]
winrate	Double	Percentage of matches won

Preprocessing of dataset

- We utilized the scikitLearn built in StandardScaler function for preprocessing
- This is important because it scales all of our data to the same range, preventing larger numeric values from biasing the data
- This is especially important for KNN, since it uses distance to calculate the closest neighbors

StandardScaler formula:

$$z = (x - u) / s$$

u = mean of samples

s = standard deviation

x = standard score of a sample

Research Targets

1

**Predicting world rank
solely based on Auto
and Endgame EPA**

2

**Predicting win rate
based on Auto, Coral,
and Barge Ranking
Points**

3

**Optimizing
predictability of win
rate and rank through
alternative distance
formulas**

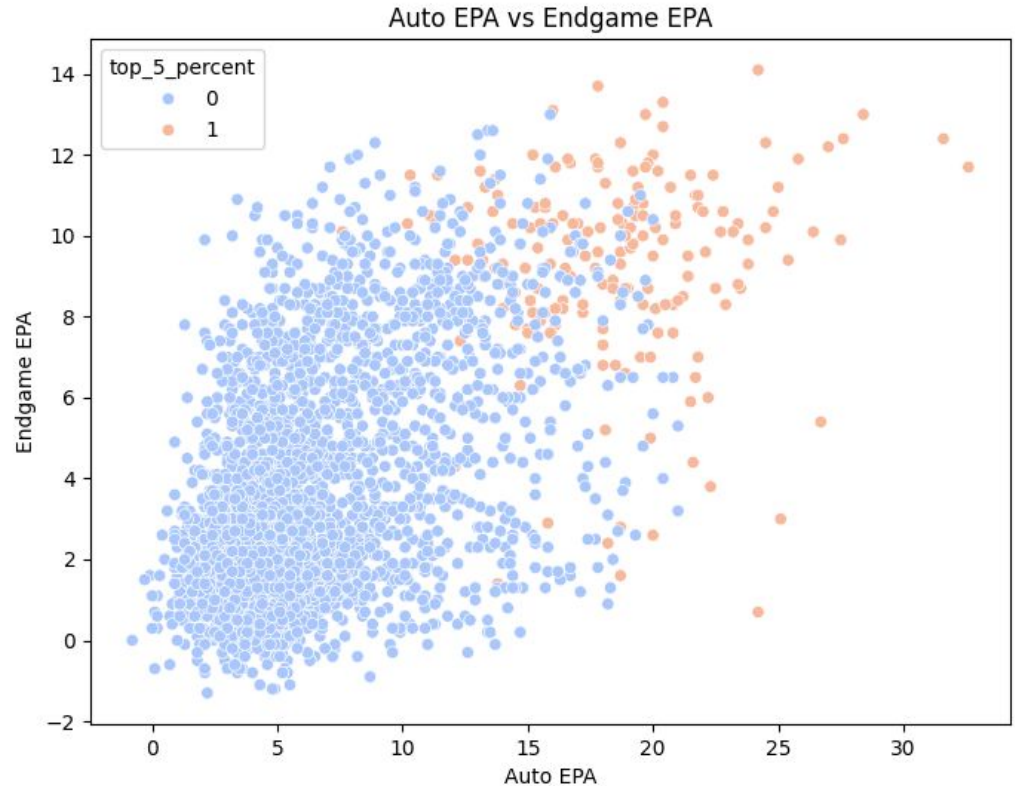
Question 1

Is it possible to predict if a team is in the **top 5% in the world** (based on EPA rank) based solely on **Endgame EPA** and **Autonomous EPA**?

Binary classification:
in top 5% or in bottom 95%

Natural clusters - top 5 near top right

KNN *could* be a valid approach due to noticeable visual separation between the clusters



k-Optimization

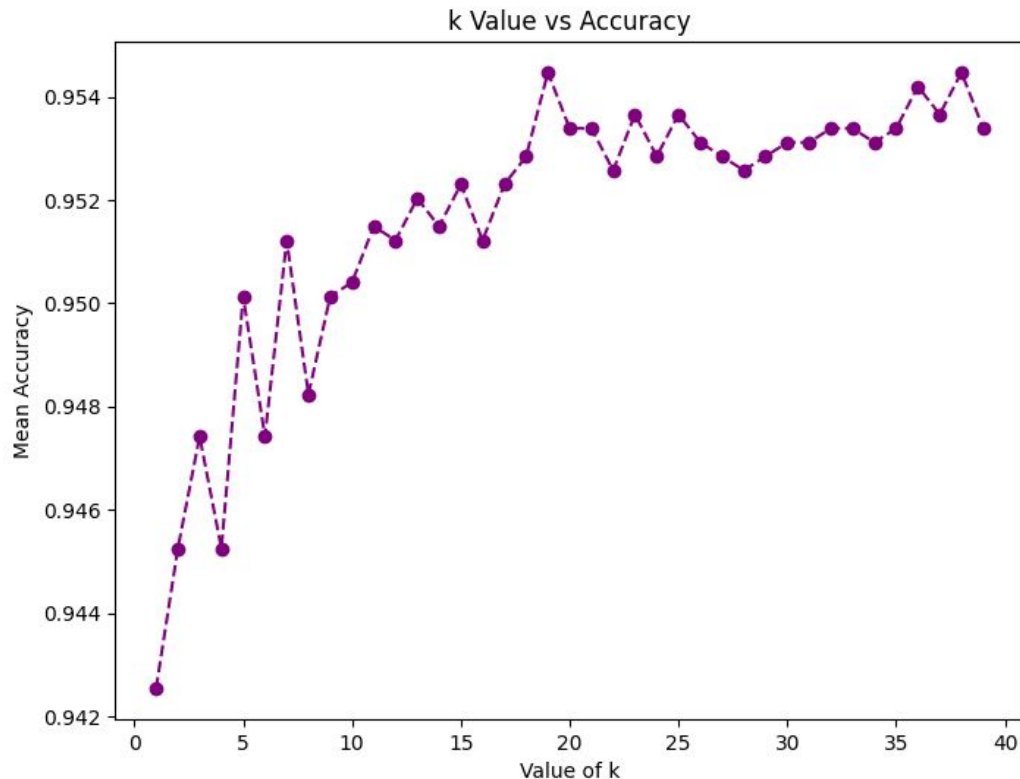
10-fold cross validation (buckets)
to optimize k

- Less biased resulting k
- Tests from multiple angles

Data standardized
(StandardScaler)

Ran KNN for $1 \leq k \leq 40$

Picked **k = 19**



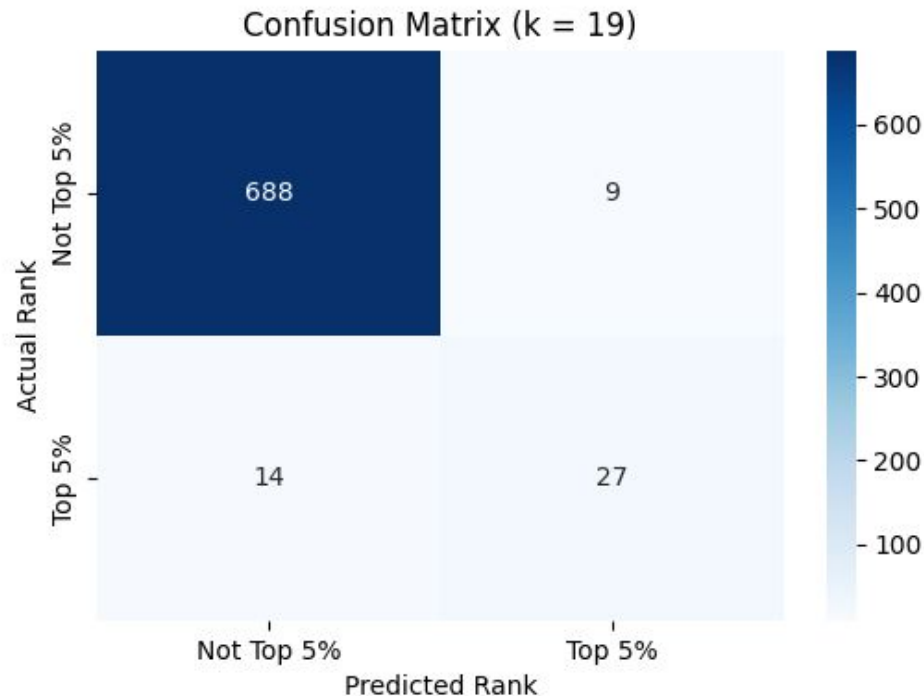
Predicting on test data, cross reference

738 unseen teams used to cross reference

27 elite teams, 14 false negatives (TII)
688 standard teams, 9 false positives (T1)

Model statistically biased to guess "Not top 5%" because of the huge 95/5 split.

Statistically more probable for there to be a higher rate of false negatives.



Question 2

How do **Auto RP**, **Coral RP**, and **Barge RP** predict the win rate of a team?

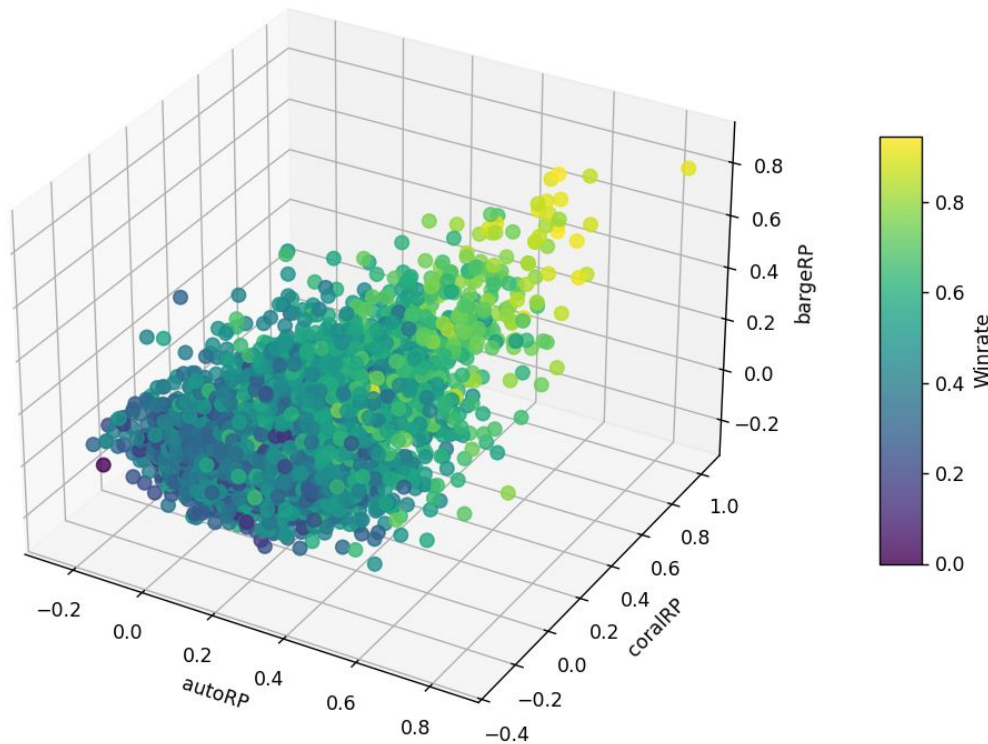
Regression KNN model

Winrate: Percentage

Natural gradient – The higher winrates are in the regions with higher RP EPAs

The autoRP seems to have less of an effect on the winrate than the other two RPs

Expected Winrate dependent on EPA



Best k: 104

Best cross-validated MSE: 0.01176197647021977

k-Optimization

10-fold cross validation
(buckets) to optimize k

- Less biased resulting k
- Tests from multiple angles

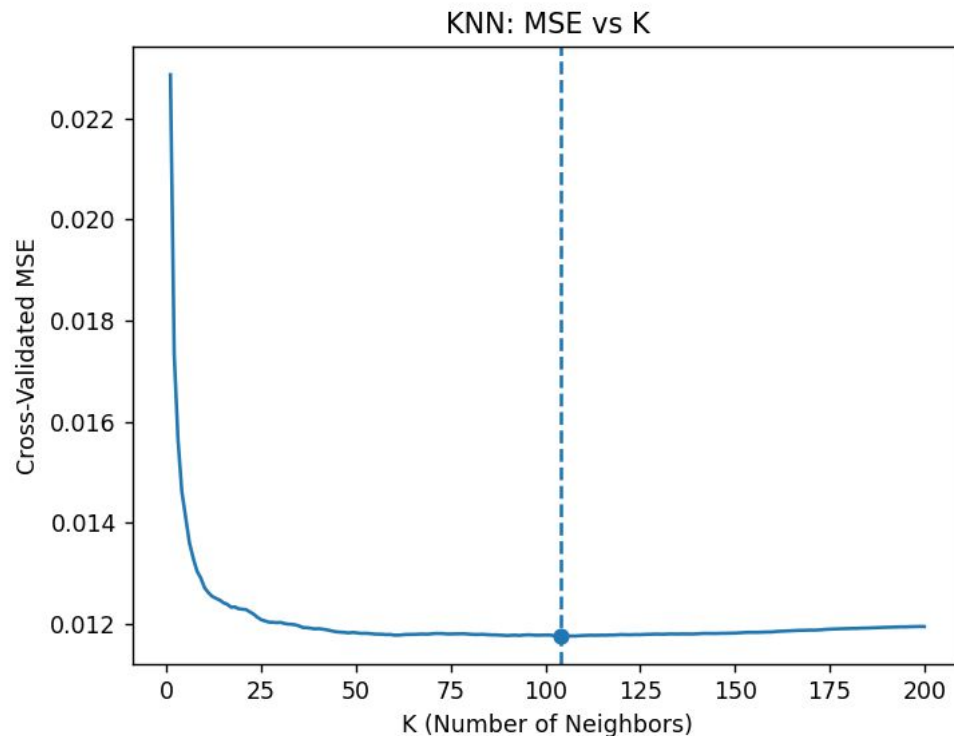
Data standardized
(StandardScaler)

Ran KNN for $1 \leq k \leq 200$

Picked **k = 104**

Error drops significantly before
104

Upward trend in error after 104



Predicting using sample data

Plugged in the values for both of our teams into the cross validated KNN model using the data from statbotics

Percent error for Roboeagles = 20.24%

Percent error for Hitchhikers = 26.86%

This could be due to the fact that there are different districts in FRC, and winrates hold different values in each region

Roboeagles:

AutoRP: 0.23

CoralRP: -0.02

BargeRP: -0.03

Predicted Winrate: 0.461

Actual winrate: 0.578

Hitchhikers:

AutoRP: 0.35

CoralRP: 0.04

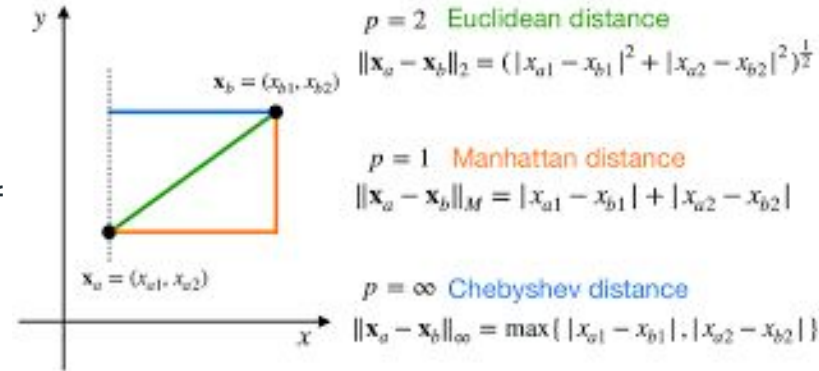
BargeRP: 0.54

Predicted Winrate: 0.482

Actual winrate: 0.659

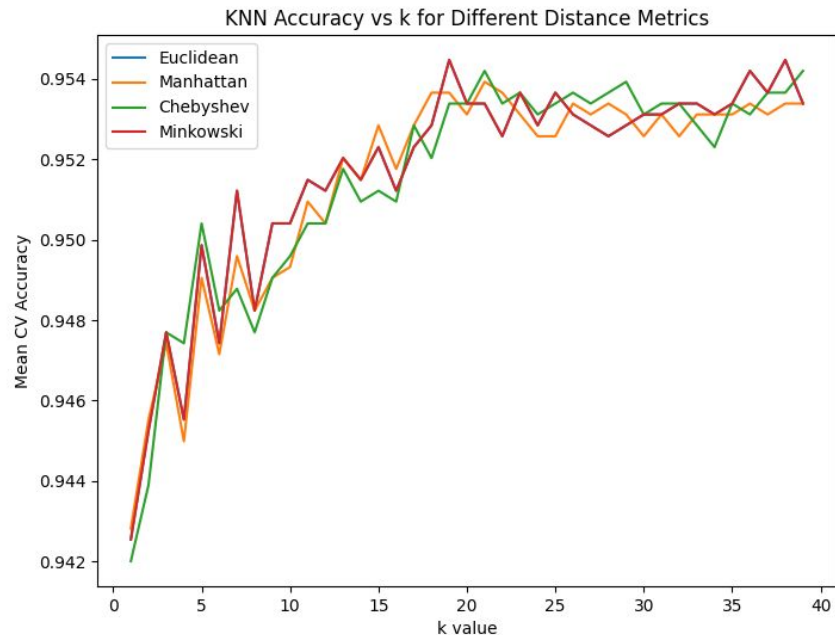
Question 3

- Can different distance calculations be used to optimize the accuracy of the prediction model?
- Euclidean: best when magnitude is important and data is normalized
- Chebyshev: good for high dimensional data (calculates max distance) but influenced by outliers
- Manhattan: good for data with outliers and lots of dimensions

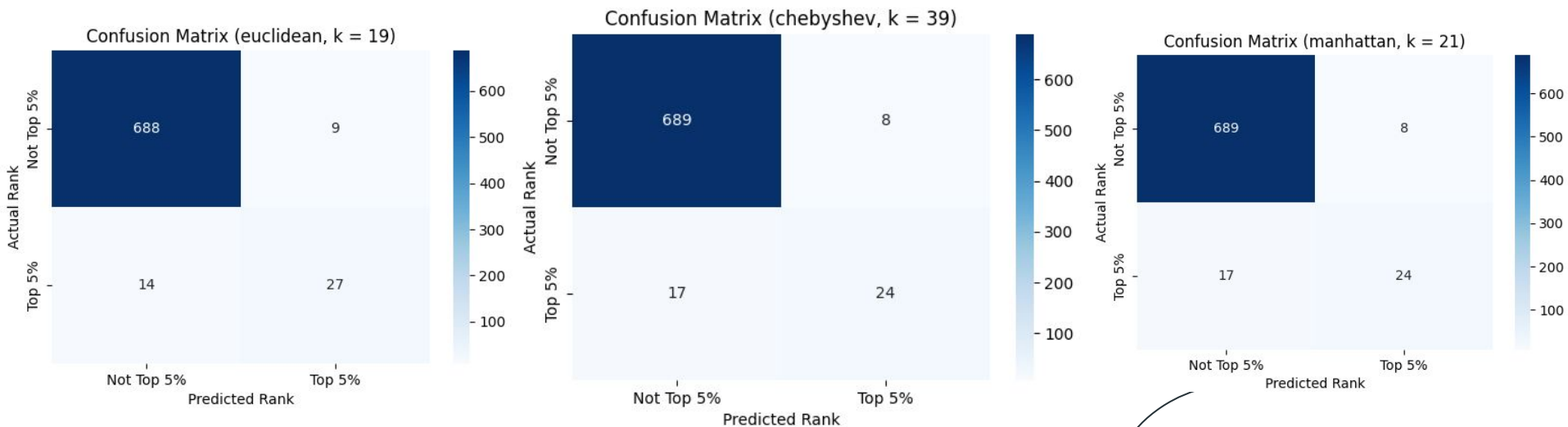


Distance Calculations in Classification

- Manhattan: $k = 21$, Mean Accuracy = 95.39%
- Chebyshev: $k = 39$, Mean Accuracy = 95.42%
- Euclidean: $k = 19$, Mean Accuracy = 95.45%
- Running with 10 folds (buckets) and scaling to eliminate bias in distance calculations



Confusion Matrices



Distance Calculations in Regression

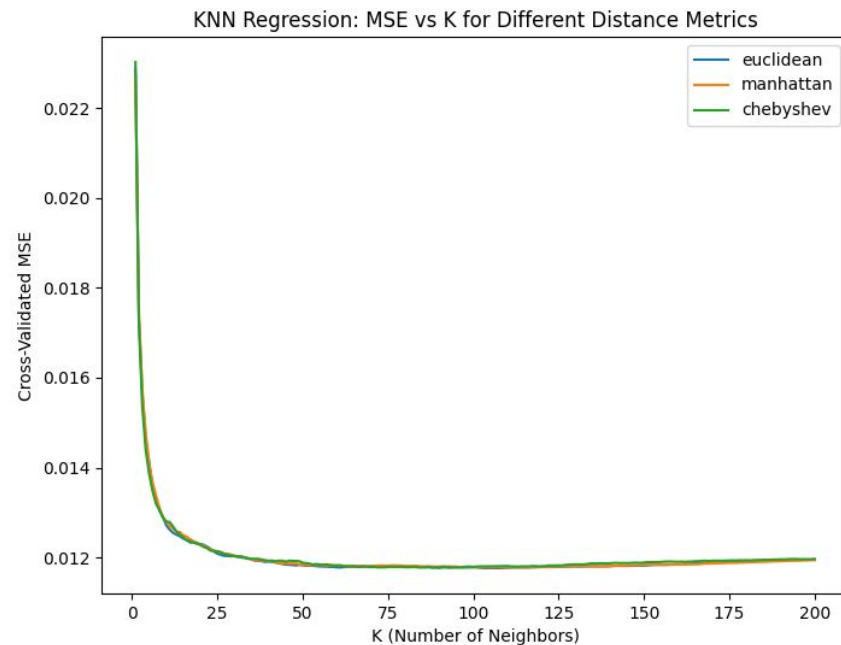
Euclidean: best $k = 104$, $MSE = 0.011762$

Manhattan: best $k = 110$, $MSE = 0.011779$

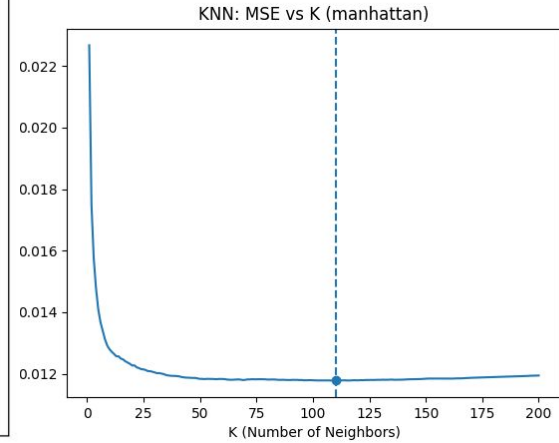
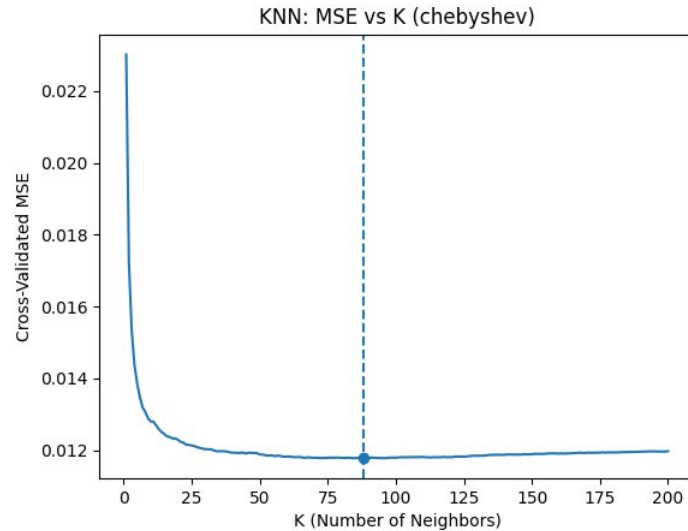
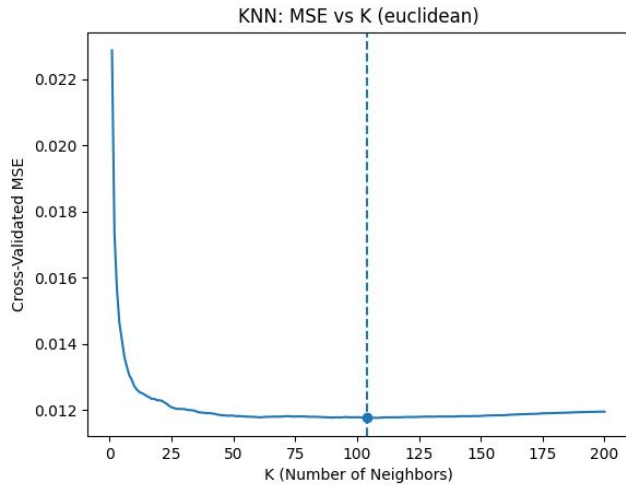
Chebyshev: best $k = 88$, $MSE = 0.117713$

MSE: Mean squared error, squared difference between predicted and actual value

Lower is better



Separated Graphs of MSE vs. K



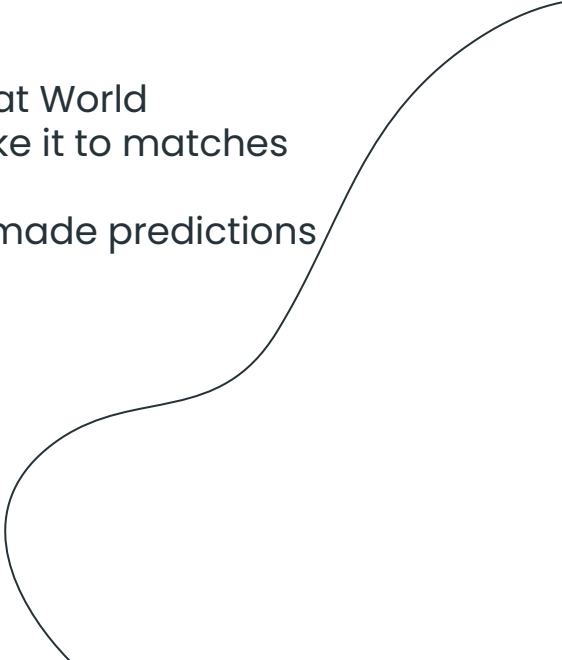
Testing Distances in Regression

- Roboeagles Prediction Error
 - Euclidean: 26.6%
 - Manhattan: 25.1%
 - Chebyshev: 25.8%
- Hitchhikers Prediction Error
 - Euclidean: 16.99%
 - Manhattan: 16.99%
 - Chebyshev: 16.8%

```
Roboeagles Predictions:  
euclidean: 0.424  
manhattan: 0.433  
chebyshev: 0.429  
Actual winrate: 0.578
```

```
Hitchhikers Predictions:  
euclidean: 0.547  
manhattan: 0.547  
chebyshev: 0.548  
Actual winrate: 0.659
```

Critique of Data Sets

- Understanding how Ranking Point EPA is calculated
 - Large number of outliers could have skewed data
 - Top teams in the world (consistently make final field at World Championship) vs. teams that don't consistently make it to matches
 - Teams sorted/separated by region or district might have made predictions more accurate
- 

Summary

- Using KNN for scouting purposes has flaws
- Due to the vast differences between districts, using a general data set may not be accurate
- In FRC, the gap between top teams and the vast majority of the rest of the field is very high
- Just using RPs is not a good metric, since some teams build specifically for those RPs and not necessarily to win
- Using alternative distance formulas yields very similar results, you cannot use different ones to optimize the model
- However, combining these models with traditional scouting methods could provide value



**Thanks for
listening!**